EA Repository Workbench: Setup Guide

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Overview

Setting up a model takes about two minutes on a Windows PC with Excel: copy the folder, allow code import once, run the installer. This guide is for whoever installs and looks after the Workbench. Architects who only use it need the User Guide.

The EA Repository Workbench is an enterprise architecture repository built on Excel, with an ArchiMate 4 metamodel. One folder holds one model:

Item	What it is
EA_Core.xlsm	The repository: metamodel (MM_ sheets), master data (DATA_Master, DATA_Relationships), history (H_ sheets), review and system sheets, and the engine code
EA_Entities.xlsm	One editable sheet per entity type (T_ sheets), with expandable relationship rows
EA_Reports.xlsx	Macro-free reports, created by Refresh reports; refreshes itself from EA_Core
model.json	Model ID, name, parent model, schema version
src\	The engine as text files, imported by the installer
History\	Version snapshots, merge bases, archived history
Templates\ Imports\ Exports\	Import templates, filled templates waiting to import, exports
Library\	draw.io libraries: ArchiMate4.xml and ModelLibrary.xml
Diagrams\	draw.io diagrams; Diagrams\Export\ holds exported images
Reports\	The Power BI project (EA_Reports.pbip)
Merge\	In a child model: the merge package for its parent

The Workbench_Tools folder holds the EA Workbench Setup window (Setup.bat), which creates new models, installs them into Excel and runs the self-tests. It needs Python and is only for the person who sets models up, not for users.

Requirements

Only Windows desktop Excel is required; draw.io and Power BI are free and optional.

Component	Needed for	Version
Windows 10 or 11	Everything (the macros use Windows components)	Any current build
Microsoft Excel for Windows, desktop	The repository, forms, sync, import, export	Excel 2016 or later; Microsoft 365 recommended
Microsoft 365 or Excel 2021	EA_Reports.xlsx dashboard formulas (dynamic arrays)	Current channel
draw.io Desktop	Diagrams, harvest, image export	Current release, installed per user or for all users
Power BI Desktop	The Power BI project (optional)	Current release
A local folder	Every model folder	Not OneDrive, SharePoint or a network path that Excel opens as a web address

The macros use components built into Windows: the Scripting runtime, MSXML 6 (reading and writing diagrams) and the Windows Script Host (draw.io image export). Regex validation rules use the VBScript regular-expression engine; where it is missing, those rules are skipped and Validate All says so.

On Mac or Excel on the web the workbooks open for reading only; no macros run.

Install a model

A model folder arrives as two .xlsx workbooks plus the src folder; the installer turns them into macro workbooks. Do this once per model folder.

With the Setup window (recommended). Double-click `Workbench_Tools\Setup.bat`, put the model folder in step 3, click **Unblock downloaded files**, then **Install into Excel**. The window opens Excel, imports the installer and starts it; you confirm in Excel. You still tick *Trust access to the VBA project object model* yourself (step 3 below) before installing; the window tells you if it is off. The manual steps below do the same by hand.

1. **Copy the folder to a local drive**, for example `C:\EA\EA_Model_Sample`. Do not work from a zip preview or a synced cloud folder.
2. **Unblock downloaded files.** If the folder came from email or the internet, right-click each .xlsx, choose Properties, tick **Unblock**, and click OK.
3. **Allow code import (once per PC).** In Excel: File > Options > Trust Center > Trust Center Settings > Macro Settings > tick **Trust access to the VBA project object model**. You can untick it again after setup; it is only needed while the installer runs.
4. **Open EA_Core.xlsx**, press **Alt+F11** to open the VBA editor, then File > Import File and pick `src\modInstall.bas`.
5. **Run the installer.** Press **F5**, choose **InstallWorkbench** and confirm.
6. **Add a trusted location (recommended).** Trust Center > Trusted Locations > Add new location, pick the folder that holds your model folders, and tick *Subfolders of this location are also trusted*. Macros then run without a yellow warning bar.

The installer imports the engine modules, saves both workbooks as .xlsm, moves the original .xlsx files to History, regenerates the type sheets, adds the Home buttons and runs Validate All. It takes under a minute for the sample model.

If Excel shows **Enable Content** when you open a workbook later, click it. Without macros the workbooks still open, but nothing is checked or synced.

First run checks

Five checks confirm the install; each takes under a minute. The acceptance test files TEST_Phase1.md to TEST_Phase5.md in the model folder go deeper.

- [ ] **Open EA_Entities.xlsm.** It opens EA_Core by itself and runs a quick sync. The Home sheet shows the buttons and the entity counts; the sample has 14 entities and 12 relationships.
- [ ] **Validate All** (Home). The sample reports no errors.
- [ ] **Sync Now** (Home). It finishes with no differences to review, and rebuilds the two draw.io libraries.
- [ ] **Refresh reports** (EA_Core Home). EA_Reports.xlsx is created next to EA_Core and opens on its Dashboard. If Excel asks about privacy levels, choose to ignore them for this file.
- [ ] **draw.io libraries.** In draw.io Desktop: File > Open Library from > Device, pick `Library\ArchiMate4.xml`, then again for `Library\ModelLibrary.xml`. Open `Diagrams\Claims_Overview.drawio` to see a generated diagram.

For Power BI, open `Reports\EA_Reports.pbip`, set the **ModelFolder** parameter to the model folder (ending with a backslash) and refresh; `Reports\README_PowerBI.md` describes the report pages to lay out.

Start a new model

New models are created with the **EA Workbench Setup** window: double-click `Workbench_Tools\Setup.bat`. It needs Python 3.9 or later on the machine that creates models (from python.org, with *Add python.exe to PATH* ticked); people who only use a model need nothing but Excel.

1. **Check this computer.** The window runs the check when it opens: Python version, the openpyxl and pytest packages, the Workbench code, Excel and draw.io. If a package is missing, click **Install Python packages**.
2. **Create a new model.** Pick the folder to create it in, type a model name such as *Group Landscape*, and tick *Include the sample data* only for a training model. The window shows the folder name it will use (`EA_Model_Group_Landscape`) and builds the workbooks with the ArchiMate 4 metamodel, the Power BI project, the draw.io libraries, the code, the guides and the test sheets.
3. **Install into Excel.** The new folder is filled in under step 3; click **Install into Excel** (see Install a model).

The model name becomes the folder name, and the folder name seeds the model ID, so give every model its own name. To start from part of an existing model instead, use **Create child** in that model (User Guide, Parent and child models).

`Setup (menu).bat` offers the same choices as a numbered menu in a console window, for computers where the window cannot open. The scripts can also be run directly; `Workbench_Tools\README.md` lists the commands.

Upgrade a model

A new engine version is installed by running the installer again over the existing model; the data stays where it is.

1. Open the model and click **Version history > Take snapshot**, so there is a copy to go back to.
2. Replace the model's `src` folder with the new one.
3. Open EA_Core.xlsm, press Alt+F11, import the new `src\modInstall.bas` (it replaces the old one), and run **InstallWorkbench**. Trust access to the VBA project object model must be on while it runs.

The installer swaps every module, keeps the .xlsm files, regenerates the type sheets and runs Validate All. The version installed is in SYS_Settings under **WorkbenchVersion**; this release is 1.1.

Models built with a pre-release of the Workbench (before 1.0) lack system sheets added later, such as SYS_Views and MERGE_Review, and there is no automatic migration for them. Start production models from the 1.1 Setup window.

Settings reference

Settings live in the SYS_Settings sheet of EA_Core. Only DrawioPath and ProtectPassword are meant to be edited by hand; the Workbench maintains the rest.

Key	Meaning	Edit?
ModelID	Unique ID of this model; checked by imports, diagrams and merges	No
ModelName	Display name, used in file names of snapshots and exports	Yes, carefully
SchemaVersion	Metamodel version; raised by every metamodel change	No
CoreFile, EntitiesFile, ReportsFile	File names of the workbooks; do not rename the files	No
ProtectPassword	Password of the protected sheets (default ea-workbench)	Yes
DrawioPath	draw.io.exe, if it is not in a standard install location	Yes
WorkbenchVersion	Engine version	No
LastValidateAll, LastSync, LastFullSync, WBSaveStamp	Housekeeping for sync and validation	No
ParentModelID, ParentModelName, ParentFolder, ChildCreated	Set in a child model: its parent and when it was created	No

The draw.io program is found automatically in the per-user location (`%LOCALAPPDATA%\Programs\draw.io`) and in Program Files; set DrawioPath only for another location.

Troubleshooting

Most problems come from macros being off, the folder being in the wrong place, or two models open at once.

Message or symptom	Cause	Fix
Excel is blocking access to the VBA project	Trust access to the VBA project object model is off	Turn it on (Install a model, step 3), run InstallWorkbench again
Yellow bar: macros have been disabled	The folder is not trusted	Click Enable Content, or add a trusted location
This workbook is open from OneDrive or SharePoint	The folder is synced or opened from the web	Copy the model folder to a local folder such as C:\EA
Another model is open	Excel cannot open two workbooks with the same name (every model has an EA_Core.xlsm)	Close the other model's workbooks, then open this one
EA_Core was not found next to this workbook	EA_Entities was moved or opened from a copy	Keep both workbooks in the same folder; do not rename them
Cells turn red or amber	A validation rule failed (red = error, amber = warning)	Hover over the cell for the message; fix the value
A row keeps coming back after deleting it	Deleting a row does not delete an entity	Set its Status to Retired
Type sheets were not refreshed	Unsaved edits or open items on SYNC_Review	Save EA_Entities, settle SYNC_Review, run Sync Now
Page is saved compressed	draw.io saved the diagram compressed	In draw.io untick File > Properties > Compressed, save
draw.io Desktop was not found	Installed in another place	Set DrawioPath in SYS_Settings to draw.io.exe
Excel asks about privacy levels (reports)	Power Query combines two sources	Choose to ignore privacy levels for this file
Snapshot failed	Disk full, or the folder is read-only	Free space or fix folder permissions, then retry

Message or symptom	Cause	Fix
Setup.bat: Python 3 is not installed	No Python on the computer, or not on PATH	Install Python from python.org with Add python.exe to PATH ticked, then run Setup.bat again
Setup window: openpyxl MISSING	The Python package that writes Excel files is not installed	Click Install Python packages; if pip is blocked, ask IT to install openpyxl

If something else stops with a message, note the exact text and what you clicked; a snapshot is taken before every risky operation, so Version history can put the model back.